

CMPG 325 — COMPUTER NETWORKS

INDIVIDUAL SEMESTER PROJECT — MILESTONE 3

TECHNICAL REPORT

Field	Details
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Module	CMPG 325 — Computer Networks
Project ID	CMPG325-2026-104
Client ID	CLI-104
Client	Ramotshere Moiloa Local Municipality Offices (Zeerust)
Industry	Municipal Services
Assigned Challenge	NAT (Inside/Outside Address Translation)
Design Constraint	CCTV traffic must be segmented from office data traffic
Change Request	CR4 — New application/file server accessible to authorised departments only
Addressing Block	10.40.0.0/16
Submission Date	16 October 2026

Declaration of Originality

I, Bongani Nkosi (37786458), declare that this project is my own original work. All sources of information have been appropriately cited, and any AI assistance used has been in compliance with the NWU AI Policy. I understand that I remain fully responsible for the correctness, understanding, verification, and academic integrity of everything submitted.

Signature:  Date: 16/08/2026

Milestone 3 — Final Submission

1. EXECUTIVE SUMMARY

1.1 Project Overview

This report documents the design, implementation, and testing of an enterprise-grade computer network for the **Ramotshere Moiloa Local Municipality Offices** in Zeerust, North West Province. The network was built and simulated entirely in **Cisco Packet Tracer 8.3** and addresses the connectivity, security, and service requirements specified in the project brief (CMPG325-2026-104).

1.2 Problem Statement

The municipality required a network that:

- Provides reliable connectivity across five departments (Administration, Finance, Human Resources, Public Services, and IT)
- Segments sensitive **CCTV traffic** from office data traffic for security
- Provides **NAT (Network Address Translation)** for external communication using the assigned block **10.40.0.0/16**
- Restricts access to a new **application/file server** so only authorised departments can reach it (Change Request CR4)

1.3 Solution Delivered

The final network solution includes:

- **Hierarchical three-tier design:** Core, Distribution, and Access layers

- **8 VLANs** for logical segmentation
- **Inter-VLAN routing** via a Layer 3 Core Switch
- **DHCP services** for all office departments (centralised on the Core-SW)
- **NAT (PAT + Static)** on the Edge Router to provide internet access for office users while excluding CCTV
- **ACL 101** to enforce CCTV segmentation
- **ACL 102** to enforce CR4 server access restrictions
- **Full documentation** including configs, testing evidence, and troubleshooting logs

1.4 Key Outcomes

Requirement	Status
Departmental connectivity	✓ Achieved
Inter-VLAN routing	✓ Achieved
CCTV segmentation (Constraint)	✓ Achieved
NAT (Inside/Outside Address Translation)	✓ Achieved
CR4 Server access control	✓ Achieved
DHCP for office users	✓ Achieved
Tested, working Packet Tracer implementation	✓ Achieved

1.5 Report Structure

The remainder of this report is organised as follows:

- **Section 2** provides an introduction and project scope.
- **Section 3** analyses the client requirements, constraints, and design assumptions.
- **Section 4** describes the physical and logical network design.
- **Section 5** details the implementation and device configurations.
- **Section 6** explains the assigned NAT challenge in depth.
- **Section 7** presents the testing and verification evidence.
- **Section 8** documents the troubleshooting performed during development.
- **Section 9** reflects on the project, lessons learned, and future improvements.
- **Section 10** lists references.

The **Appendix** contains full device configurations and evidence screenshots.

2. INTRODUCTION

2.1 Background

The Ramotshere Moiloa Local Municipality, based in Zeerust, operates multiple administrative departments that require reliable and secure data communications. Like many South African municipalities, it faces the dual challenge of:

1. Providing modern digital services to internal departments and the public
2. Protecting sensitive data and systems from unauthorised access

This project addresses those needs by designing and simulating a **secure, segmented, and service-rich enterprise network** using Cisco technologies.

2.2 Project Scope

In Scope

- Physical topology design (device placement, cabling)
- Logical topology design (VLANs, subnets, routing, security)
- IP addressing plan based on the assigned block **10.40.0.0/16**
- Implementation and configuration in Cisco Packet Tracer 8.3
- **Assigned challenge:** NAT (Inside/Outside Address Translation)
- **Constraint:** CCTV traffic segmentation
- **Change Request CR4:** Restricted server access for authorised departments
- Testing and verification of all functionality

Out of Scope

- Actual physical deployment
- Real hardware procurement or installation
- Production DNS or email services
- External firewall appliances (unless simulated in Packet Tracer)

2.3 Project Objectives

#	Objective
1	Design a topology appropriate for a mid-sized municipal office
2	Segment the network into logical VLANs
3	Provide inter-VLAN routing
4	Configure DHCP for office departments
5	Implement and verify NAT (PAT + Static)
6	Enforce CCTV segmentation via ACLs
7	Enforce CR4 server access via ACLs
8	Test all connectivity and security requirements
9	Document all work in a professional GitHub portfolio
10	Produce a working .pkt file, technical report, and demonstration video

2.4 Methodology

The project followed a structured engineering approach:

1. **Requirement Analysis** — Extract requirements from project brief
2. **Assumption Documentation** — Make and justify reasonable assumptions where details were unspecified
3. **Design** — Physical and logical topology, IP plan, VLAN scheme, security model
4. **Implementation** — Build and configure in Packet Tracer
5. **Testing** — Verify each requirement with evidence
6. **Documentation** — Produce report, portfolio, and video

2.5 Tools Used

Tool	Purpose
Cisco Packet Tracer 8.3	Network design and simulation

Tool	Purpose
Visual Studio Code	Markdown documentation writing
Draw.io	Topology diagrams
GitHub	Portfolio of evidence
Microsoft Word / VS Code	Report authoring (Markdown → PDF)

2.6 Assumptions Summary

The following major assumptions were made (detailed in Section 3):

- 5 main departments exist (Administration, Finance, HR, Public Services, IT)
- ~75 total users across all departments
- 8-12 CCTV cameras monitoring the premises
- Two servers (Application + File) plus a dedicated DHCP server
- Three-building campus layout

2.7 Deliverables

The final deliverables for Milestone 3 are:

1. **Working Packet Tracer file** (.pkt)
2. **GitHub portfolio** with all documentation and evidence
3. **This technical report** (Markdown → PDF)
4. **15-20 minute individual video demonstration** with webcam inset
5. **Supporting files** (configuration exports, screenshots)

3. CLIENT ANALYSIS

3.1 Client Overview

Field	Value
Organisation	Ramotshere Moiloa Local Municipality Offices

Field	Value
Location	Zeerust, North West Province, South Africa
Industry	Municipal Services
Client ID	CLI-104
Project ID	CMPG325-2026-104
Assigned Block	10.40.0.0/16

3.2 Organisational Context

The Ramotshere Moiloa Local Municipality provides essential public services to the Zeerust community. Its administrative offices manage:

- Financial records (rates, billing, procurement)
- Human resources and payroll
- Public-facing services (permits, licensing, community engagement)
- Security monitoring (CCTV across municipal buildings)

The network must serve multiple departments with distinct data, privacy, and access requirements.

3.3 Client Requirements (from Project Brief)

#	Requirement
1	Provide appropriate connectivity and network services
2	Accommodate the stated design constraint (CCTV segmentation)
3	Address Change Request CR4 (restricted server access)
4	Implement the assigned challenge: NAT (Inside/Outside Address Translation)
5	Use the assigned addressing block: 10.40.0.0/16
6	Produce a working, testable Packet Tracer implementation

3.4 Design Constraint

"CCTV traffic must be segmented from office data traffic."

This is a **non-negotiable security requirement**. CCTV cameras handle sensitive video streams and must not be reachable from office VLANs, and vice versa. This requires:

- A dedicated CCTV VLAN
- ACL enforcement on the CCTV gateway interface
- CCTV devices to be excluded from NAT (no internet access)

3.5 Change Request CR4

"A new application/file server is installed and must be reachable by authorised departments only."

This change request adds a **server access control layer**:

- Only **Administration, Finance, and IT** departments can reach the servers
- **HR and Public Services** are denied
- **CCTV** is also denied (in line with the segmentation constraint)

3.6 Assumptions

Because the project brief does not specify all organisational details, the following **reasonable engineering assumptions** were made and justified.

3.6.1 Departmental Structure

Department	Assumed Users	Justification
Administration	15	Executive, legal, registry
Finance	12	Budget, payroll, procurement
Human Resources	8	Staff records, recruitment
Public Services	20	Customer-facing services
IT/Technical	6	Network and systems support

Department	Assumed Users	Justification
Security/CCTV	4	Monitoring operators
Total	65 users	Mid-sized municipal office

3.6.2 Device Count

Device Type	Quantity	Justification
PCs	~12 (simulated)	One per department minimum, scaled down for simulation
Printers	5 (1 per department)	Typical municipal setup
Servers	3	Application, File, DHCP
CCTV Cameras	8-12	Standard perimeter + internal coverage
Switches	6	Core + 5 access
Routers	2	Edge + ISP simulation

3.6.3 Building Layout

Assumed **three-building campus**:

- 1. **Main Admin Building** — Admin, Finance, HR, IT
- 2. **Public Services Centre** — Public-facing services
- 3. **IT/Server Room** — Core infrastructure + servers

Plus a **separate CCTV building/zone** for camera operations.

3.6.4 Services Required

Service	Reason
DHCP	Simplify client management
Inter-VLAN routing	Enable departmental communication

Service	Reason
NAT (PAT)	Internet access for office staff
Static NAT	Server-side external reachability
ACLs	Security enforcement
DNS (public)	External name resolution

3.6.5 Internet Connectivity

Assumed:

- Single ISP uplink
- Public IP range **192.168.1.0/30** for external facing
- Public IP pool **192.168.1.2 - 192.168.1.4** for NAT translations

3.7 Justification of Design Decisions

Decision	Justification
VLAN per department	Standard best practice; isolates broadcast domains; enables ACL policies
Layer 3 Core Switch	Centralises routing; efficient inter-VLAN traffic; scalable
Centralised DHCP on Core-SW	Simplifies management; avoids external DHCP relay complexity
Separate server VLAN (200)	Enables precise ACL control for CR4
Static IPs for servers	Required for reliable NAT static mappings and services
DHCP for office PCs	Reduces admin overhead
CCTV VLAN isolated	Enforces security constraint
ACL 101 for CCTV	Prevents CCTV ↔ office traffic
ACL 102 with bidirectional rules	Extended ACLs are stateless; both directions must be permitted for authorised flows

3.8 Requirements Traceability Matrix

Requirement	Design Element	Section
Connectivity	VLANs, trunk links, inter-VLAN routing	4
NAT (assigned challenge)	PAT + Static NAT on Edge Router	5, 6
CCTV Segmentation	VLAN 100 + ACL 101	5, 6
CR4 Server Access	VLAN 200 + ACL 102	5, 6
DHCP	Core-SW DHCP pools	5
Addressing Block 10.40.0.0/16	VLSM subnet allocation	4
Packet Tracer Implementation	Working .pkt file	Appendix A

4. NETWORK DESIGN

4.1 Design Overview

The network follows a **hierarchical three-tier architecture**:

- **Core Layer** — High-speed Layer 3 switching and inter-VLAN routing
- **Distribution Layer** — (Merged with Core in this design for scale)
- **Access Layer** — Departmental switches providing user connectivity

This design balances cost, performance, and manageability — appropriate for a mid-sized municipal network.

4.2 Physical Topology

4.2.1 Device Inventory

Device	Model	Quantity	Role
ISP Router	Cisco ISR 4321	1	Simulated ISP uplink

Device	Model	Quantity	Role
Edge Router	Cisco ISR 4321	1	NAT Gateway, routing
Core Switch	Cisco 3650-24PS (L3)	1	Inter-VLAN routing, DHCP, ACLs
Access Switches	Cisco 2960-24TT	5	Department access
Servers	Server-PT	3	App, File, DHCP
PCs	PC-PT	12	End users
CCTV Cameras	PC-PT (simulated cameras)	2	CCTV endpoints

4.2.2 Physical Layout

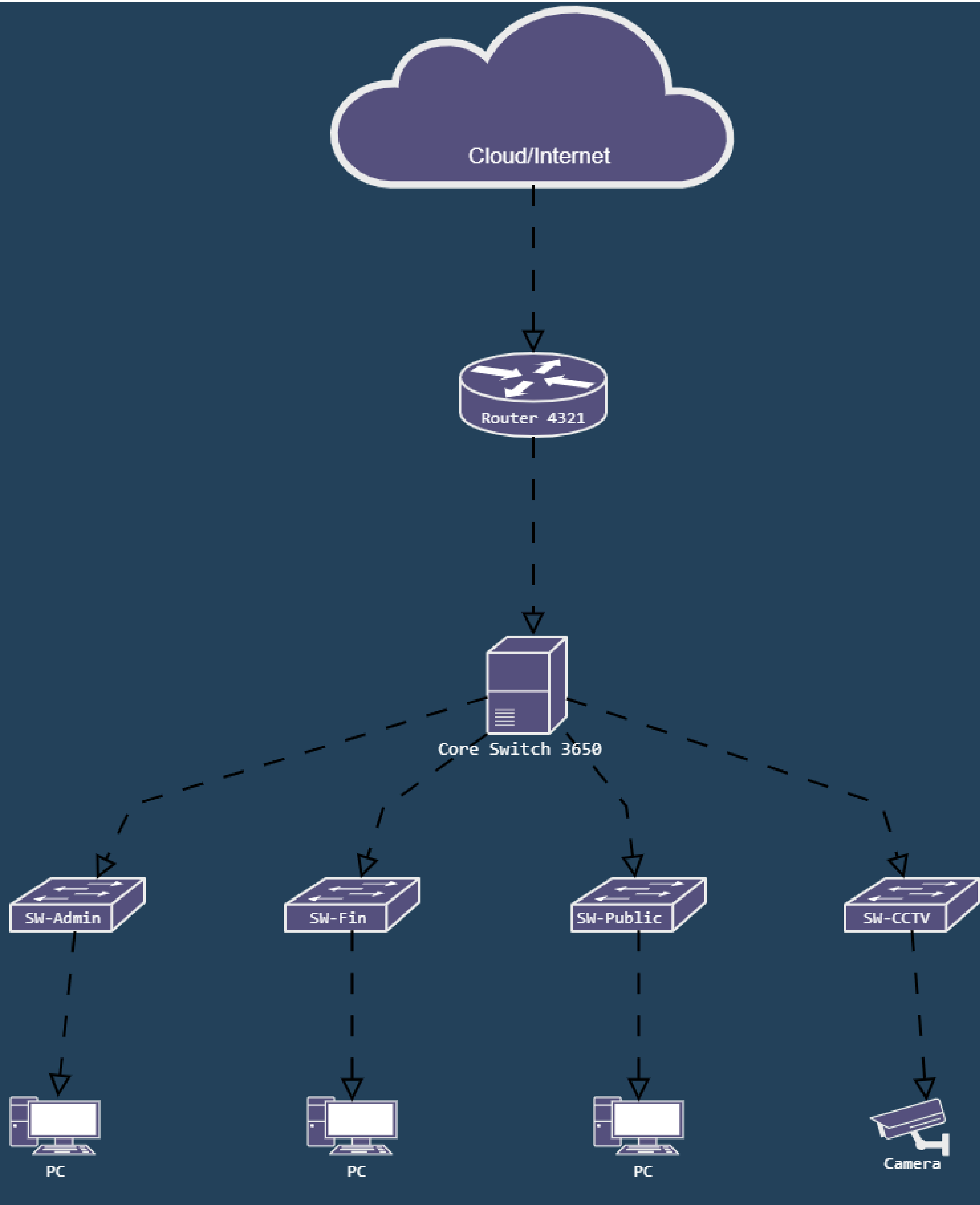
Three-building campus layout:

Building	Departments	Access Switch
Main Admin Building	Admin, Finance, HR, IT	SW-Admin, SW-Fin, SW-HR
Public Services Centre	Public Services	SW-Public
CCTV Building	Security cameras	SW-CCTV
IT/Server Room	Core infrastructure	(Core-SW, Edge-Router, Servers)

4.2.3 Cabling

Link Type	Medium	Speed
Core ↔ Edge Router	Copper Straight-Through	1 Gbps
Core ↔ Access Switches	Copper Cross-Over	100 Mbps
Access ↔ End Devices	Copper Straight-Through	100 Mbps
Router ↔ ISP Router	Copper Straight-Through	1 Gbps

4.2.4 Physical Diagram



4.3 Logical Topology

4.3.1 VLAN Design

VLAN	Name	Subnet	Gateway	Purpose
10	ADMIN	10.40.1.0/24	10.40.1.254	Administration department
20	FINANCE	10.40.2.0/24	10.40.2.254	Finance department
30	HR	10.40.3.0/24	10.40.3.254	Human Resources
40	PUBLIC	10.40.4.0/24	10.40.4.254	Public Services
50	IT	10.40.5.0/24	10.40.5.254	IT Department (reserved)
100	CCTV	10.40.10.0/24	10.40.10.254	CCTV (Segmented)
200	SERVERS	10.40.20.0/28	10.40.20.14	Server farm
999	MANAGEMENT	10.40.255.0/28	10.40.255.254	Network management

Note: VLAN 50 (IT) is reserved. In the current implementation, IT PCs are connected via SW-Admin and reside in VLAN 10 (see Section 8 for details and future improvement).

4.3.2 VLSM Summary

The assigned block **10.40.0.0/16** (65,536 addresses) is divided using VLSM:

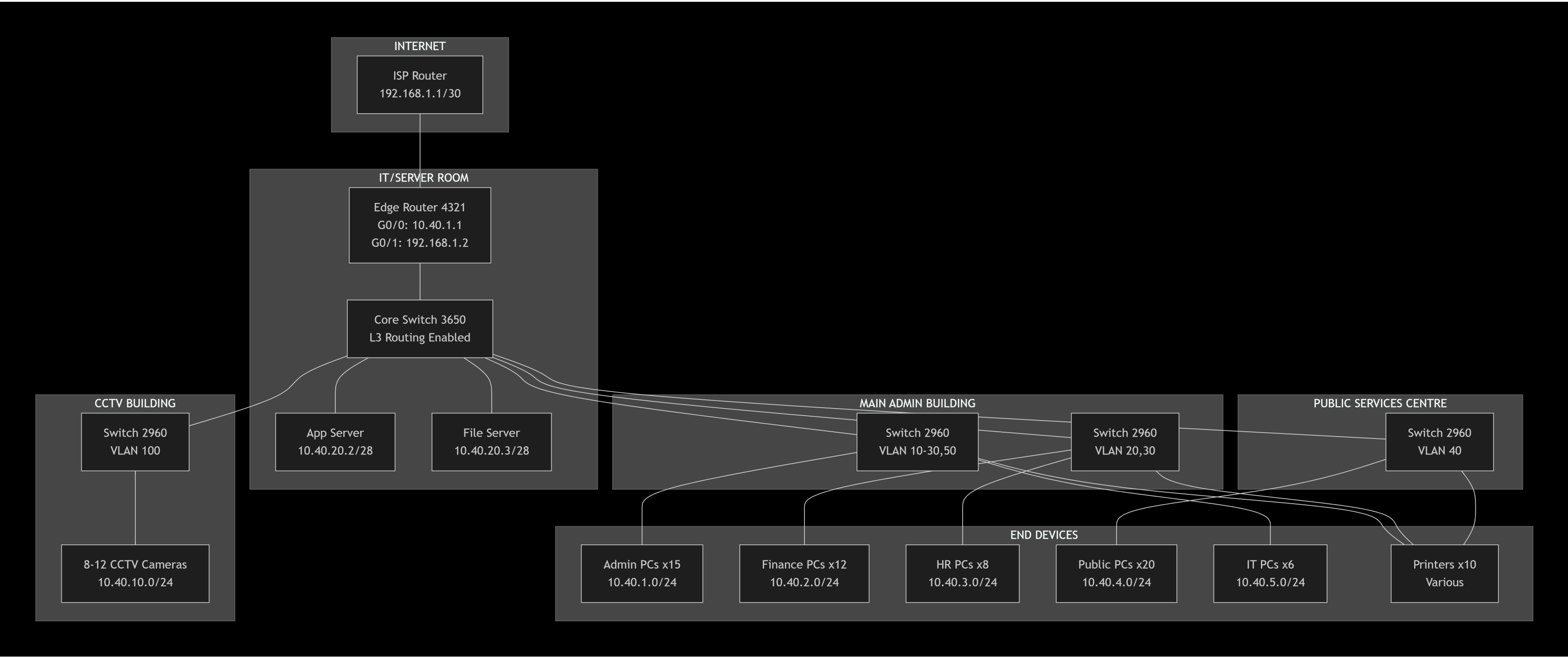
Subnet	Range	Mask	Size	Usage
10.40.1.0/24	.1.1 - .1.254	/24	254	Admin
10.40.2.0/24	.2.1 - .2.254	/24	254	Finance
10.40.3.0/24	.3.1 - .3.254	/24	254	HR
10.40.4.0/24	.4.1 - .4.254	/24	254	Public
10.40.5.0/24	.5.1 - .5.254	/24	254	IT
10.40.10.0/24	.10.1 - .10.254	/24	254	CCTV
10.40.20.0/28	.20.1 - .20.14	/28	14	Servers

Subnet	Range	Mask	Size	Usage
10.40.255.0/28	.255.1 - .255.14	/28	14	Management
10.40.0.0/16 remainder	—	/16	Large	Reserved for expansion

4.3.3 IP Addressing Plan — Key Devices

Device	Interface	IP	Mask
Edge-Router	G0/0/0 (Inside)	10.40.1.1	/24
Edge-Router	G0/0/1 (Outside)	192.168.1.2	/30
Core-SW	VLAN 10	10.40.1.254	/24
Core-SW	VLAN 20	10.40.2.254	/24
Core-SW	VLAN 30	10.40.3.254	/24
Core-SW	VLAN 40	10.40.4.254	/24
Core-SW	VLAN 50	10.40.5.254	/24
Core-SW	VLAN 100	10.40.10.254	/24
Core-SW	VLAN 200	10.40.20.14	/28
Core-SW	VLAN 999	10.40.255.254	/28
App-Server	NIC	10.40.20.2	/28
File-Server	NIC	10.40.20.3	/28
DHCP-Server	NIC	10.40.20.4	/28

4.3.4 Logical Diagram



4.4 Routing Design

4.4.1 Routing Method

- **Inter-VLAN routing** performed by Core-SW (Layer 3)
- **Static routing** between Core-SW and Edge-Router
- **Default route** on Edge-Router to ISP

4.4.2 Routing Table (Core-SW)

Route	Type	Next Hop
10.40.1.0/24	Connected (Vlan10)	—
10.40.2.0/24	Connected (Vlan20)	—
10.40.3.0/24	Connected (Vlan30)	—
10.40.4.0/24	Connected (Vlan40)	—
10.40.5.0/24	Connected (Vlan50)	—
10.40.10.0/24	Connected (Vlan100)	—
10.40.20.0/28	Connected (Vlan200)	—

Route	Type	Next Hop
10.40.255.240/28	Connected (Vlan999)	—
0.0.0.0/0	Static (default)	10.40.1.1

4.4.3 Routing Table (Edge-Router)

Route	Type	Next Hop
10.40.1.0/24	Connected	—
192.168.1.0/30	Connected	—
10.40.0.0/16	Static	10.40.1.254
0.0.0.0/0	Static (default)	192.168.1.1

4.5 NAT Design

4.5.1 NAT Types Used

Type	Purpose
PAT (Overload)	Many office PCs → single public IP (192.168.1.2)
Static NAT	Server public mappings (192.168.1.3, 192.168.1.4)

4.5.2 NAT Mappings

Inside Local	Inside Global	Purpose
10.40.1.0/24	192.168.1.2	PAT (dynamic)
10.40.2.0/24	192.168.1.2	PAT (dynamic)
10.40.3.0/24	192.168.1.2	PAT (dynamic)
10.40.4.0/24	192.168.1.2	PAT (dynamic)
10.40.5.0/24	192.168.1.2	PAT (dynamic)

Inside Local	Inside Global	Purpose
10.40.20.2	192.168.1.3	Static (App Server)
10.40.20.3	192.168.1.4	Static (File Server)
10.40.10.0/24	NONE (excluded)	CCTV excluded by design

4.5.3 NAT Interfaces

Interface	Role
Edge-Router G0/0/0	ip nat inside
Edge-Router G0/0/1	ip nat outside

4.6 Security Design

4.6.1 Security Layering

Layer	Control
VLAN segmentation	Isolates departments
ACL 101	Blocks CCTV → office traffic
ACL 102	Restricts server access to Admin/Finance/IT
NAT exclusion	CCTV has no internet access
Password protection	<code>enable secret cisco123</code> on all network devices

4.6.2 ACL Design

ACL 101 — CCTV Segmentation

Extended IP access list 101

```
10 deny ip 10.40.10.0 0.0.0.255 10.40.0.0 0.0.255.255
```

```
20 permit ip any any
```

Applied to: `interface vlan 100 (in)`

ACL 102 — CR4 Server Access

```
Extended IP access list 102
10 permit ip 10.40.1.0 0.0.0.255 10.40.20.0 0.0.0.15
20 permit ip 10.40.2.0 0.0.0.255 10.40.20.0 0.0.0.15
30 permit ip 10.40.5.0 0.0.0.255 10.40.20.0 0.0.0.15
40 permit ip 10.40.20.0 0.0.0.15 10.40.1.0 0.0.0.255
50 permit ip 10.40.20.0 0.0.0.15 10.40.2.0 0.0.0.255
60 permit ip 10.40.20.0 0.0.0.15 10.40.5.0 0.0.0.255
70 permit ip 10.40.20.0 0.0.0.15 10.40.20.0 0.0.0.15
80 deny ip any any
```

Applied to: interface vlan 200 (in)

Why bidirectional rules?

Extended ACLs are **stateless** — they don't track connections. To allow both requests and replies, both directions must be explicitly permitted. This is a deliberate design decision documented in Section 8.

4.7 Services Design

4.7.1 DHCP

Centralised on Core-SW, providing IPs for VLANs 10, 20, 30, 40, 50.

Pool	Network	Gateway	DNS
VLAN10_POOL	10.40.1.0/24	10.40.1.254	8.8.8.8
VLAN20_POOL	10.40.2.0/24	10.40.2.254	8.8.8.8
VLAN30_POOL	10.40.3.0/24	10.40.3.254	8.8.8.8
VLAN40_POOL	10.40.4.0/24	10.40.4.254	8.8.8.8
VLAN50_POOL	10.40.5.0/24	10.40.5.254	8.8.8.8

4.7.2 Servers

Server	IP	Service
App-Server	10.40.20.2	HTTP/HTTPS
File-Server	10.40.20.3	FTP
DHCP-Server	10.40.20.4	(Backup / repurposed)

4.8 Design Decisions Summary

Decision	Rationale
L3 Core-SW for routing	Centralised, scalable, cost-effective
DHCP on Core-SW	Simplicity, no relay complexity
Servers in dedicated VLAN	Precise ACL control for CR4
Bidirectional ACL 102	Stateless ACLs require explicit return permits
CCTV VLAN excluded from NAT ACL	Enforces no-internet security requirement
Static IPs for servers	Reliable NAT mappings + service hosting

End of Section 4

5. IMPLEMENTATION

5.1 Implementation Overview

All devices were configured in **Cisco Packet Tracer 8.3** using IOS CLI commands. This section documents the key configurations applied to each device in the network.

5.2 Device Configuration Summary

Device	Hostname	Key Configurations
Edge Router	Edge-Router	NAT (PAT + Static), routing, NAT inside/outside interfaces
Core Switch	Core-SW	VLANs, SVIs, inter-VLAN routing, DHCP pools, ACLs
Access Switches	SW-Admin, SW-Fin, SW-HR, SW-Public, SW-CCTV	VLANs, trunk ports, access ports
Servers	App-Server, File-Server, DHCP-Server	Static IPs, service configurations
PCs	Multiple	DHCP clients
CCTV Cameras	CCTV-Cam1, CCTV-Cam2	Static IPs in VLAN 100

5.3 Edge-Router Configuration

The Edge-Router performs the **NAT (Inside/Outside Address Translation)** — the assigned challenge for this project.

5.3.1 Interfaces

```
interface GigabitEthernet0/0/0
description INSIDE-LAN
ip address 10.40.1.1 255.255.255.0
ip nat inside
no shutdown

interface GigabitEthernet0/0/1
description OUTSIDE-ISP
ip address 192.168.1.2 255.255.255.252
ip nat outside
no shutdown
```

5.3.2 Routing

```
ip route 0.0.0.0 0.0.0.0 192.168.1.1  
ip route 10.40.0.0 255.255.0.0 10.40.1.254
```

5.3.3 NAT Configuration

ACL for NAT (excludes CCTV):

```
access-list 1 permit 10.40.1.0 0.0.0.255  
access-list 1 permit 10.40.2.0 0.0.0.255  
access-list 1 permit 10.40.3.0 0.0.0.255  
access-list 1 permit 10.40.4.0 0.0.0.255  
access-list 1 permit 10.40.5.0 0.0.0.255  
access-list 1 permit 10.40.20.0 0.0.0.15
```

PAT (Overload):

```
ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
```

Static NAT (CR4 Servers):

```
ip nat inside source static 10.40.20.2 192.168.1.3  
ip nat inside source static 10.40.20.3 192.168.1.4
```

5.4 Core-SW Configuration

The Core-SW is the heart of the network — providing inter-VLAN routing, DHCP, and ACL enforcement.

5.4.1 IP Routing Enabled

```
ip routing
```

5.4.2 VLANs Created

```
vlan 10  
name ADMIN  
vlan 20  
name FINANCE  
vlan 30  
name HR
```

```
vlan 40
name PUBLIC
vlan 50
name IT
vlan 100
name CCTV
vlan 200
name SERVERS
vlan 999
name MANAGEMENT
```

5.4.3 SVIs (Gateways)

```
interface Vlan10
ip address 10.40.1.254 255.255.255.0
no shutdown
```

```
interface Vlan20
ip address 10.40.2.254 255.255.255.0
no shutdown
```

```
interface Vlan30
ip address 10.40.3.254 255.255.255.0
no shutdown
```

```
interface Vlan40
ip address 10.40.4.254 255.255.255.0
no shutdown
```

```
interface Vlan50
ip address 10.40.5.254 255.255.255.0
no shutdown
```

```
interface Vlan100
ip address 10.40.10.254 255.255.255.0
no shutdown
```

```
interface Vlan200
ip address 10.40.20.14 255.255.255.240
no shutdown
```

```
interface Vlan999
ip address 10.40.255.254 255.255.255.240
no shutdown
```

5.4.4 Trunk Ports to Access Switches

```
interface FastEthernet0/5
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,50,200
```

```
interface FastEthernet0/6
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 20
```

```
interface FastEthernet0/7
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 30
```

```
interface FastEthernet0/8
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 40
```

```
interface FastEthernet0/9
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100
```

5.4.5 DHCP Pools

```
ip dhcp pool VLAN10_POOL
network 10.40.1.0 255.255.255.0
default-router 10.40.1.254
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN20_POOL
network 10.40.2.0 255.255.255.0
```

```
default-router 10.40.2.254
dns-server 8.8.8.8

ip dhcp pool VLAN30_POOL
network 10.40.3.0 255.255.255.0
default-router 10.40.3.254
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN40_POOL
network 10.40.4.0 255.255.255.0
default-router 10.40.4.254
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN50_POOL
network 10.40.5.0 255.255.255.0
default-router 10.40.5.254
dns-server 8.8.8.8
```

5.4.6 Default Route

```
ip route 0.0.0.0 0.0.0.0 10.40.1.1
```

5.4.7 ACLs

ACL 101 (CCTV Segmentation):

```
access-list 101 deny ip 10.40.10.0 0.0.0.255 10.40.0.0 0.0.255.255
access-list 101 permit ip any any
```

```
interface Vlan100
ip access-group 101 in
```

ACL 102 (CR4 Server Access):

```
access-list 102 permit ip 10.40.1.0 0.0.0.255 10.40.20.0 0.0.0.15
access-list 102 permit ip 10.40.2.0 0.0.0.255 10.40.20.0 0.0.0.15
access-list 102 permit ip 10.40.5.0 0.0.0.255 10.40.20.0 0.0.0.15
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.1.0 0.0.0.255
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.2.0 0.0.0.255
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.5.0 0.0.0.255
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.20.0 0.0.0.15
access-list 102 deny ip any any
```

```
interface Vlan200
ip access-group 102 in
```

5.5 Access Switch Configuration

Each access switch was configured with:

- A dedicated VLAN for its department
- A trunk port connecting to Core-SW
- Access ports for end devices with `portfast` enabled

5.5.1 SW-Admin (Admin department)

```
vlan 10
name ADMIN
```

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 10
spanning-tree portfast
```

```
interface FastEthernet0/3
switchport mode access
switchport access vlan 10
spanning-tree portfast
```

```
interface FastEthernet0/4
switchport mode access
switchport access vlan 10
spanning-tree portfast
```

```
interface FastEthernet0/5
switchport mode access
switchport access vlan 10
spanning-tree portfast
```

```
interface GigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,50,200
```

5.5.2 SW-Fin (Finance department)

vlan 20

name FINANCE

interface FastEthernet0/1

switchport mode trunk

switchport trunk allowed vlan 20

switchport trunk native vlan 1

interface FastEthernet0/2

switchport mode access

switchport access vlan 20

spanning-tree portfast

interface FastEthernet0/3

switchport mode access

switchport access vlan 20

spanning-tree portfast

interface FastEthernet0/4

switchport mode access

switchport access vlan 20

spanning-tree portfast

5.5.3 SW-HR (Human Resources)

vlan 30

name HR

interface FastEthernet0/1

switchport mode trunk

switchport trunk allowed vlan 30

switchport trunk native vlan 1

interface range FastEthernet0/2-4

switchport mode access

switchport access vlan 30

spanning-tree portfast

5.5.4 SW-Public (Public Services)

vlan 40

name PUBLIC

interface FastEthernet0/1

switchport mode trunk

switchport trunk allowed vlan 40

switchport trunk native vlan 1

interface range FastEthernet0/2-4

switchport mode access

switchport access vlan 40

spanning-tree portfast

5.5.5 SW-CCTV (CCTV — Segmented)

vlan 100

name CCTV

interface FastEthernet0/1

switchport mode trunk

switchport trunk allowed vlan 100

switchport trunk native vlan 1

interface range FastEthernet0/2-4

switchport mode access

switchport access vlan 100

spanning-tree portfast

5.6 Server Configuration

5.6.1 App-Server

IP Address: 10.40.20.2

Subnet Mask: 255.255.255.240

Default Gateway: 10.40.20.14

DNS Server: 8.8.8.8

HTTP/HTTPS Services: Enabled

5.6.2 File-Server

IP Address: 10.40.20.3
Subnet Mask: 255.255.255.240
Default Gateway: 10.40.20.14
FTP Service: Enabled (user: admin / password: admin123)

5.6.3 DHCP-Server

IP Address: 10.40.20.4
Subnet Mask: 255.255.255.240
Default Gateway: 10.40.20.14
DHCP Service: Disabled (Core-SW handles DHCP)

5.7 End Device Configuration

5.7.1 Office PCs

All office PCs (Admin, Finance, HR, Public Services, IT) are configured as **DHCP clients**.

Expected IP ranges:

Department	VLAN	Expected Range
Admin	10	10.40.1.x
Finance	20	10.40.2.x
HR	30	10.40.3.x
Public	40	10.40.4.x
IT	10	10.40.1.x

5.7.2 CCTV Cameras

CCTV cameras are configured with **static IPs** in VLAN 100:

Device	IP	Mask	Gateway
CCTV-Cam1	10.40.10.10	255.255.255.0	10.40.10.254

Device	IP	Mask	Gateway
CCTV-Cam2	10.40.10.11	255.255.255.0	10.40.10.254

5.8 Implementation Verification

After all configurations were applied, the following commands were used to verify correct operation:

Command	Purpose
<code>show ip interface brief</code>	Verify IP addresses on all interfaces
<code>show vlan brief</code>	Verify VLANs and port assignments
<code>show interfaces trunk</code>	Verify trunk ports
<code>show ip route</code>	Verify routing tables
<code>show ip nat translations</code>	Verify NAT mappings
<code>show ip nat statistics</code>	Verify NAT operation
<code>show ip access-lists</code>	Verify ACL entries and match counts
<code>show ip dhcp binding</code>	Verify DHCP leases
<code>show mac address-table</code>	Verify MAC learning

All verifications were successful — see Section 7 for detailed evidence.

End of Section 5

6. ASSIGNED CHALLENGE — NAT (INSIDE/OUTSIDE ADDRESS TRANSLATION)

6.1 Challenge Overview

The assigned networking challenge for this project is:

NAT (Inside/Outside Address Translation)

NAT allows private internal IP addresses to be translated to public IP addresses for external communication. This project implements two forms of NAT:

1. **PAT (Port Address Translation / Overload)** — Many internal hosts share a single public IP
2. **Static NAT** — One-to-one permanent mapping for servers

6.2 Why NAT is Required

The Ramotshere Moiloa Local Municipality uses the **private IP block 10.40.0.0/16**, which is not routable on the public internet. Without NAT:

- Internal PCs could not reach external websites or services
- External parties could not reach the municipality's servers

NAT solves this by translating internal private IPs to a small pool of public IPs (or a single public IP with PAT).

6.3 NAT Types Explained

6.3.1 PAT (Port Address Translation)

Purpose: Allow many internal hosts to share a single public IP address by using unique source port numbers.

Example:

- Admin-PC1 (10.40.1.10) sends a packet to a website

- Edge-Router translates the source to 192.168.1.2:1024
- When the reply returns to 192.168.1.2:1024 , the router translates back to 10.40.1.10:1024

This is efficient — thousands of internal devices can share one public IP.

6.3.2 Static NAT

Purpose: Provide a stable, predictable public IP mapping for internal servers so external parties can reliably reach them.

Example:

- App-Server is at 10.40.20.2 internally
- It maps to 192.168.1.3 externally
- Any external connection to 192.168.1.3 is translated to 10.40.20.2

6.4 NAT Configuration (Edge-Router)

6.4.1 Interface Roles

NAT requires the router to know which interface faces the internal network (inside) and which faces the internet (outside).

```
interface GigabitEthernet0/0/0
description INSIDE-LAN
ip address 10.40.1.1 255.255.255.0
ip nat inside
no shutdown
```

```
interface GigabitEthernet0/0/1
description OUTSIDE-ISP
ip address 192.168.1.2 255.255.255.252
ip nat outside
no shutdown
```

Explanation:

- `ip nat inside` — traffic entering from the LAN is eligible for translation
- `ip nat outside` — traffic leaving to the ISP is translated

6.4.2 ACL for NAT (Which Traffic Gets Translated)

```
access-list 1 permit 10.40.1.0 0.0.0.255
access-list 1 permit 10.40.2.0 0.0.0.255
access-list 1 permit 10.40.3.0 0.0.0.255
access-list 1 permit 10.40.4.0 0.0.0.255
access-list 1 permit 10.40.5.0 0.0.0.255
access-list 1 permit 10.40.20.0 0.0.0.15
```

Key design decision: The CCTV subnet (10.40.10.0/24) is **deliberately excluded** from this ACL.

Effect: CCTV devices cannot be translated and therefore cannot reach the internet. This enforces the security constraint that CCTV must be isolated.

6.4.3 PAT (Overload) Configuration

```
ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
```

Breakdown:

- ip nat inside source list 1 — translates traffic matching ACL 1
- interface GigabitEthernet0/0/1 — uses the outside interface's IP (192.168.1.2)
- overload — enables PAT (port-based multiplexing) — many-to-one

6.4.4 Static NAT Configuration

```
ip nat inside source static 10.40.20.2 192.168.1.3
ip nat inside source static 10.40.20.3 192.168.1.4
```

Effect:

Inside Local	Inside Global	Purpose
10.40.20.2	192.168.1.3	App Server (CR4)
10.40.20.3	192.168.1.4	File Server (CR4)

These static entries persist permanently, allowing external access to the servers.

6.5 NAT Verification

6.5.1 Verification Command 1: `show ip nat translations`

Output observed:

```
Pro Inside global Inside local Outside local Outside global
--- 192.168.1.3 10.40.20.2 --- ---
--- 192.168.1.4 10.40.20.3 --- ---
```

Interpretation:

- Two static entries are present for the servers 
- Dynamic PAT entries appear when traffic is generated and expire after idle timeout
- The 2 static mappings confirm servers are externally reachable

6.5.2 Verification Command 2: `show ip nat statistics`

Output observed:

```
Total translations: 2 (2 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0/1
Inside Interfaces: GigabitEthernet0/0/0
Hits: 19 Misses: 32
Expired translations: 13
Dynamic mappings:
```

Interpretation:

- **2 static translations confirmed** 
- **Inside and Outside interfaces correctly assigned** 
- **Hits: 19** — NAT has actively translated traffic
- **Expired translations: 13** — Dynamic PAT entries have been created and expired (proof of active use)

6.5.3 Functional Verification — Ping Tests

Test 1: HR-PC1 → ISP-Router (192.168.1.1)

```
C:>ping 192.168.1.1
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=253
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=253
```

Reply from 192.168.1.1: bytes=32 time<1ms TTL=253
Success rate is 75 percent (3/4)

Result:  NAT working. HR-PC1 (10.40.3.x) reached a public IP via PAT.

Test 2: CCTV-Cam1 → 8.8.8.1 (Internet)

C:>ping 8.8.8.1
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Success rate is 0 percent (0/4)

Result:  CCTV correctly blocked from internet access (excluded from NAT ACL).

6.6 Why NAT Is Appropriate Here

Reason	Explanation
Conserves public IPs	Only one public IP is needed for all office users
Hides internal structure	External parties cannot see the 10.40.0.0/16 topology
Enables internet access	Office staff can reach external services
Allows external server access	Static NAT makes servers reachable
Enforces security policy	CCTV is excluded — a design decision tying to the constraint
Standard practice	Matches real-world municipal network designs

6.7 NAT Design Decisions

Decision	Rationale
PAT (Overload) for office users	Cost-effective; one public IP; sufficient for municipal scale
Static NAT for both servers	Predictable external reachability; required for CR4 server access
Exclude CCTV from NAT ACL	Reinforces the CCTV segmentation constraint

Decision	Rationale
Use a single public IP (192.168.1.2)	Simple design; scalable for the office's user count
Use Loopback for simulated internet	Optional polish — provides a pingable "internet" endpoint

6.8 Summary

The assigned NAT challenge has been:

-  **Configured** on the Edge-Router (PAT + Static NAT)
-  **Verified** using `show ip nat translations` and `show ip nat statistics`
-  **Tested** via functional ping tests (HR reaches ISP; CCTV cannot)
-  **Integrated** with the CCTV segmentation constraint (CCTV excluded from NAT)
-  **Documented** with evidence screenshots (08-nat-translations.png, 09-nat-statistics.png)

Assigned challenge successfully demonstrated.

End of Section 6

7. TESTING AND VERIFICATION

7.1 Testing Overview

All network functionality was systematically tested in Cisco Packet Tracer 8.3. Testing focused on four areas:

1. **Connectivity** — Can devices communicate as expected?
2. **NAT** — Is the assigned challenge working?
3. **CCTV Segmentation** — Is the design constraint enforced?
4. **CR4 Server Access** — Is the change request honoured?

7.2 Testing Methodology

Each test followed this structure:

- **Setup:** Source device, destination device, expected outcome
- **Command:** Exact command executed
- **Expected Result:** What should happen
- **Actual Result:** What was observed
- **Interpretation:** What this proves

Evidence was captured as screenshots and stored in the GitHub portfolio.

7.3 Test 1 — Inter-VLAN Routing (Admin → Finance)

Source: Admin-PC1 (10.40.1.x)

Destination: Fin-PC1 (10.40.2.x)

Command: `ping 10.40.2.1`

Expected: Success (inter-VLAN routing via L3 Core-SW)

Result:

Reply from 10.40.2.1: bytes=32 time<1ms TTL=127

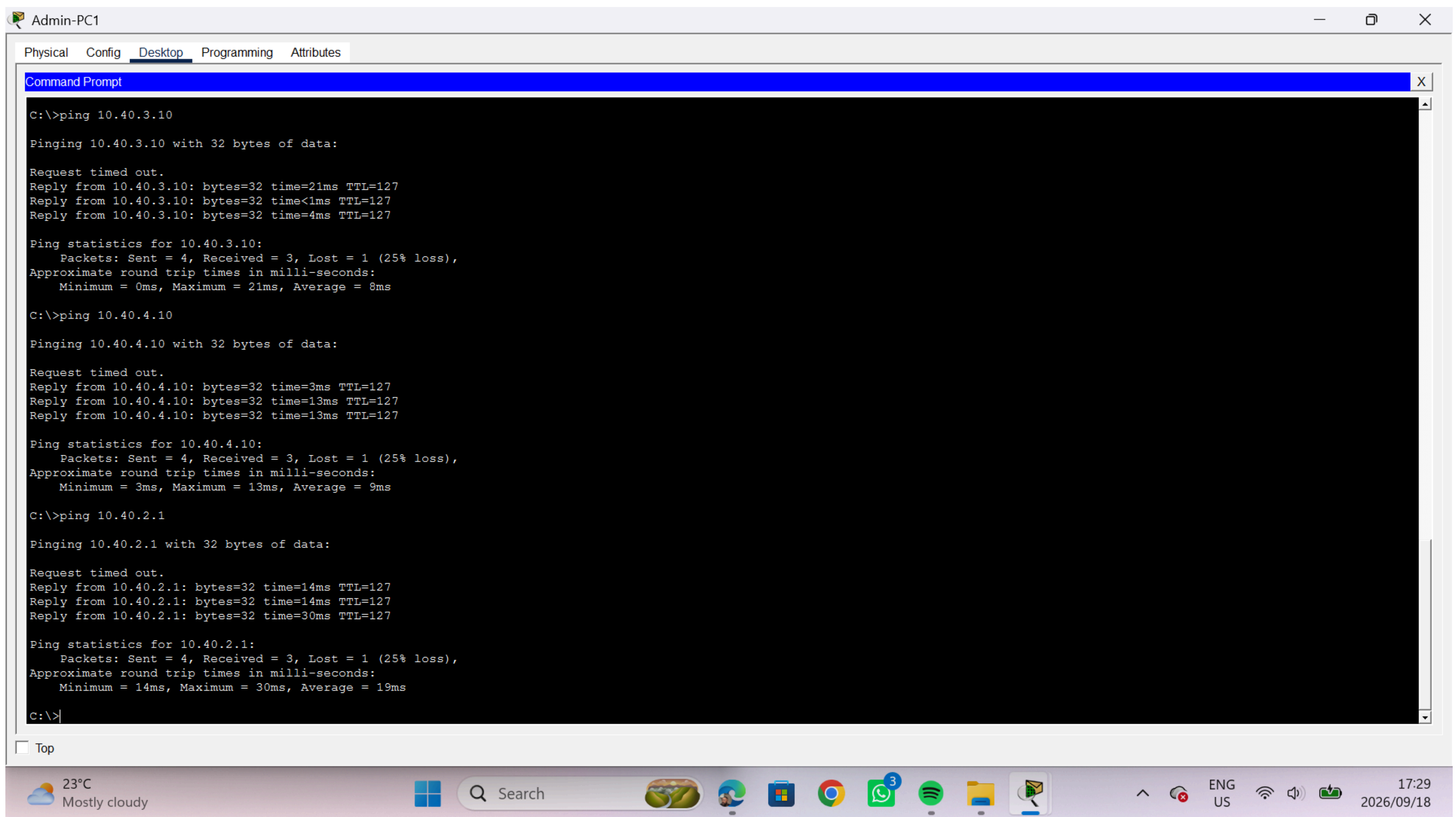
Reply from 10.40.2.1: bytes=32 time<1ms TTL=127

Reply from 10.40.2.1: bytes=32 time<1ms TTL=127

Success rate is 75 percent (3/4)

Interpretation:  Inter-VLAN routing works. The first packet is dropped due to ARP resolution — a normal behaviour.

Evidence:



7.4 Test 2 — Admin → App-Server (CR4 — Authorised)

Source: Admin-PC1 (10.40.1.x)

Destination: App-Server (10.40.20.2)

Command: ping 10.40.20.2

Expected: Success (Admin is authorised under CR4)

Result:

Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

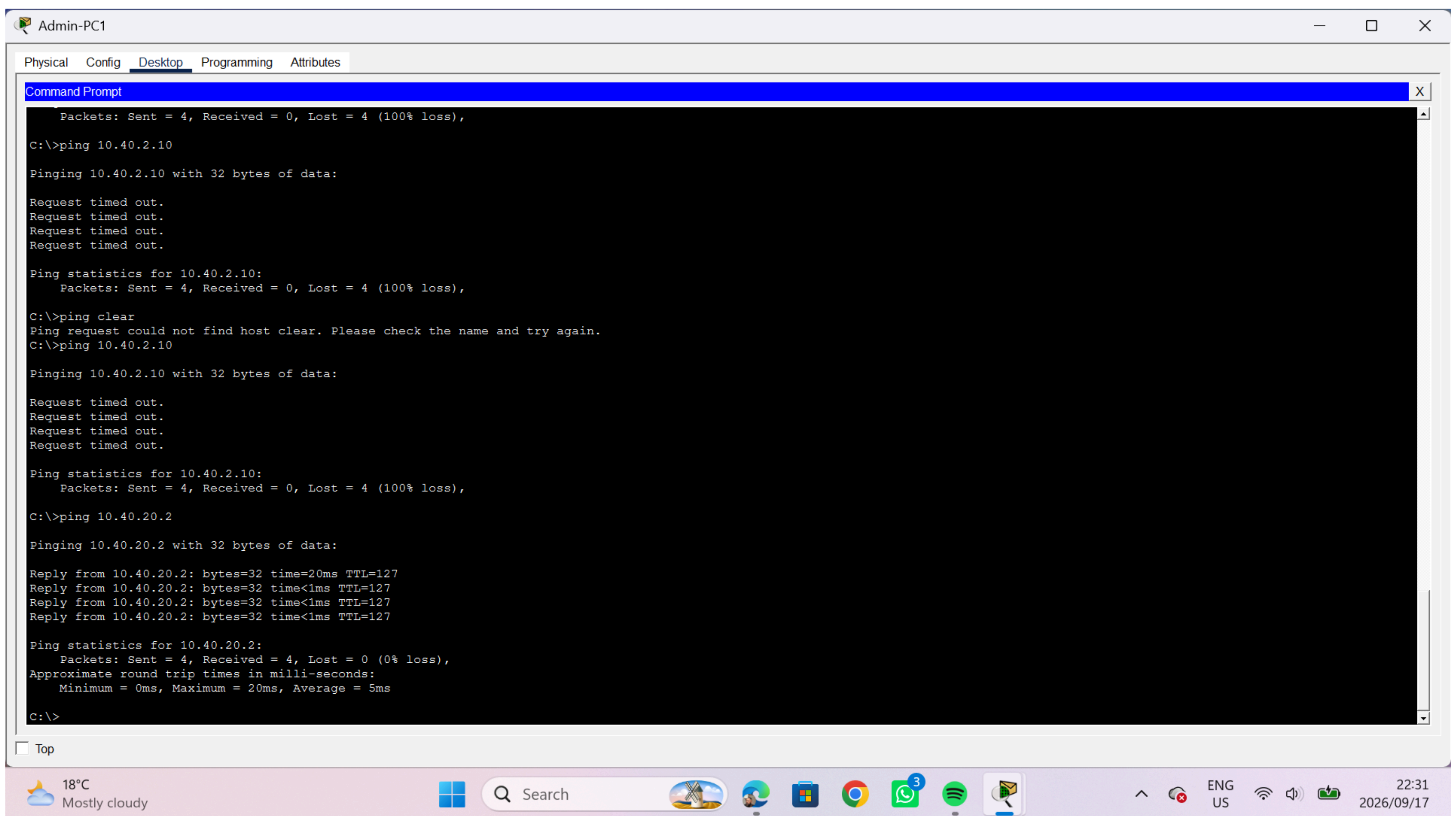
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

Reply from 10.40.20.2: bytes=32 time<1ms TTL=127

Success rate is 100 percent (4/4)

Interpretation:  ACL 102 correctly permits Admin (10.40.1.0/24) to reach the server VLAN.

Evidence:



7.5 Test 3 — HR → App-Server (CR4 — Denied)

Source: HR-PC1 (10.40.3.x)

Destination: App-Server (10.40.20.2)

Command: ping 10.40.20.2

Expected: Blocked (HR is NOT authorised under CR4)

Result:

Request timed out.

Request timed out.

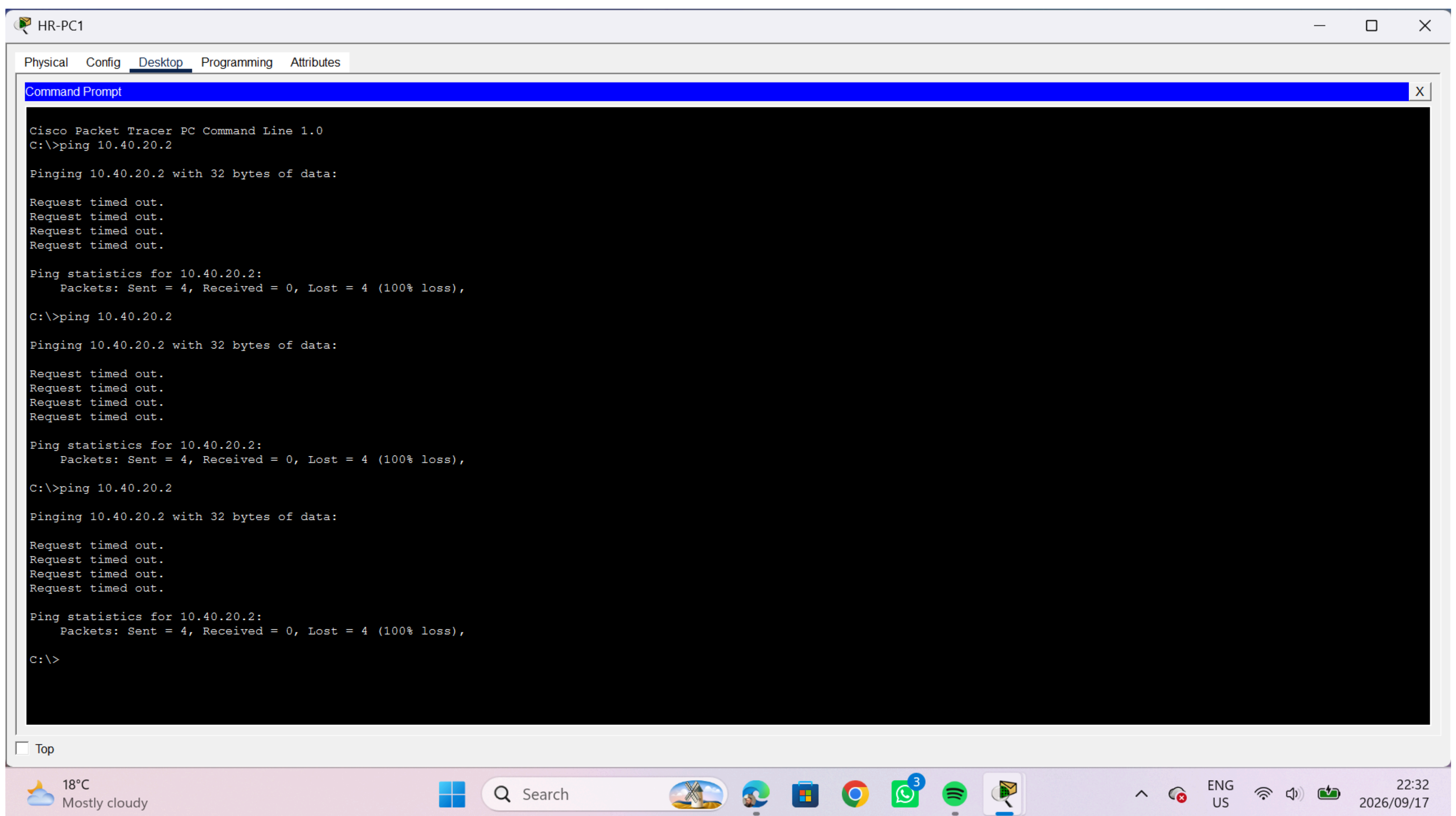
Request timed out.

Request timed out.

Success rate is 0 percent (0/4)

Interpretation:  ACL 102 correctly denies HR (10.40.3.0/24) access to the server VLAN.

Evidence:



7.6 Test 4 — HR → Internet (NAT)

Source: HR-PC1 (10.40.3.x)

Destination: ISP-Router (192.168.1.1)

Command: ping 192.168.1.1

Expected: Success (PAT translation)

Result:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=253

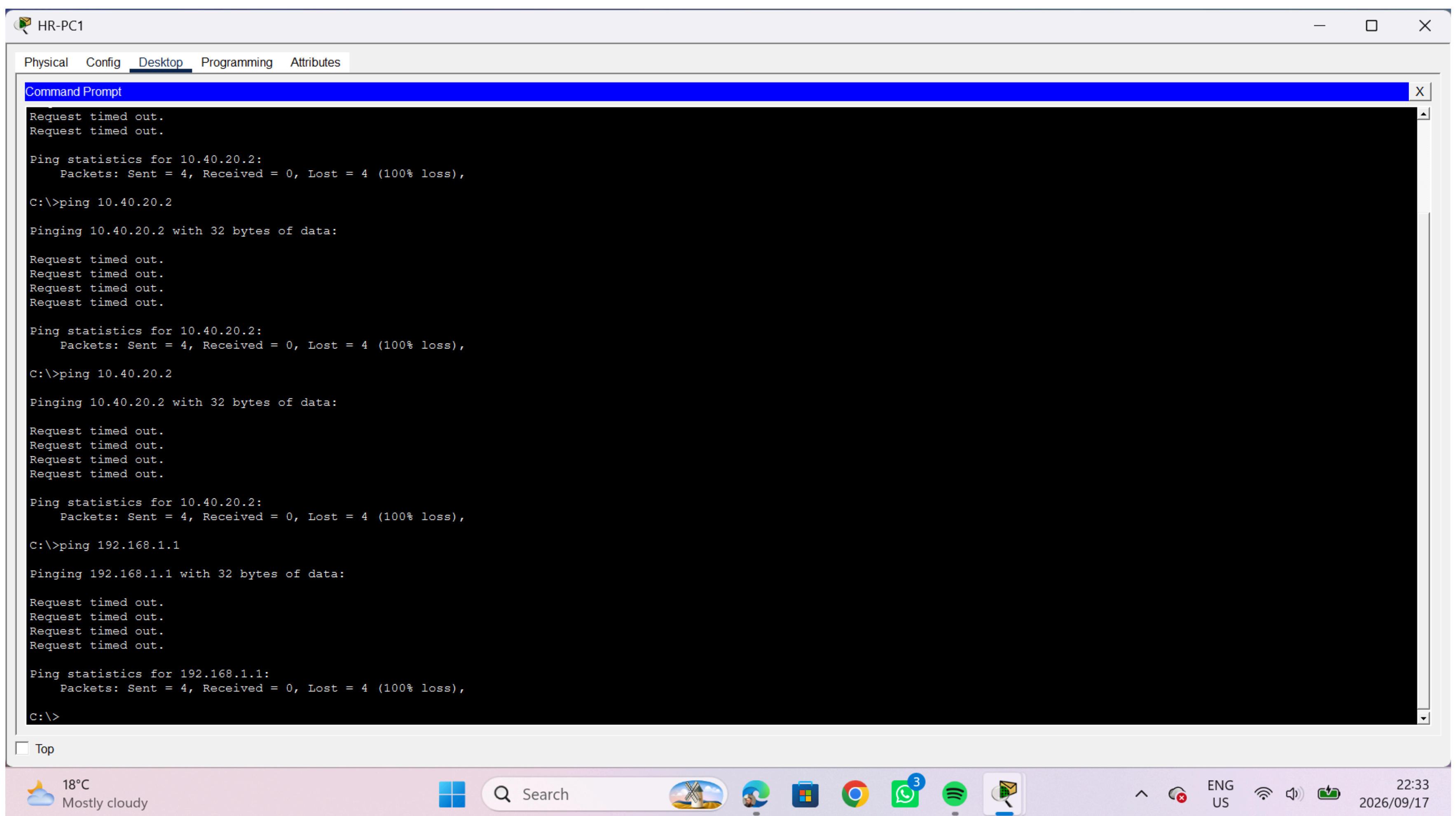
Reply from 192.168.1.1: bytes=32 time<1ms TTL=253

Reply from 192.168.1.1: bytes=32 time<1ms TTL=253

Success rate is 75 percent (3/4)

Interpretation:  PAT working. HR-PC1 (private IP) reached a public destination via NAT.

Evidence:



7.7 Test 5 — CCTV → Admin PC (Segmentation Constraint)

Source: CCTV-Cam1 (10.40.10.10)

Destination: Admin-PC1 (10.40.1.10)

Command: ping 10.40.1.10

Expected: Blocked (ACL 101 blocks CCTV → office)

Result:

Request timed out.

Reply from 10.40.10.254: Destination host unreachable.

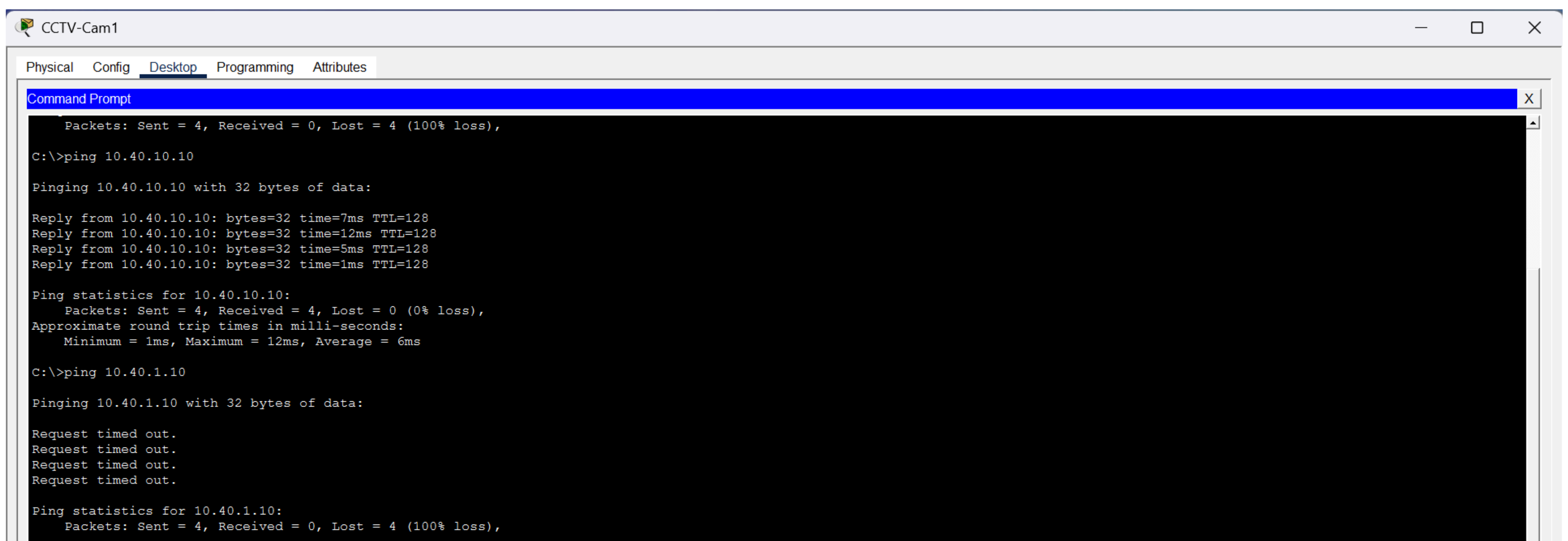
Reply from 10.40.10.254: Destination host unreachable.

Reply from 10.40.10.254: Destination host unreachable.

Success rate is 0 percent (0/4)

Interpretation:  CCTV cannot reach office VLANs. The "Destination host unreachable" message comes from the CCTV gateway (10.40.10.254), confirming ACL 101 is enforcing the block.

Evidence:



```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.40.10.10

Pinging 10.40.10.10 with 32 bytes of data:

Reply from 10.40.10.10: bytes=32 time=7ms TTL=128
Reply from 10.40.10.10: bytes=32 time=12ms TTL=128
Reply from 10.40.10.10: bytes=32 time=5ms TTL=128
Reply from 10.40.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 10.40.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 6ms

C:\>ping 10.40.1.10

Pinging 10.40.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.40.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

7.8 Test 6 — CCTV → CCTV (Internal)

Source: CCTV-Cam1 (10.40.10.10)

Destination: CCTV-Cam2 (10.40.10.10)

Command: ping 10.40.10.10

Expected: Success (same VLAN, no ACL applies)

Result:

Reply from 10.40.10.10: bytes=32 time<1ms TTL=128

Reply from 10.40.10.10: bytes=32 time<1ms TTL=128

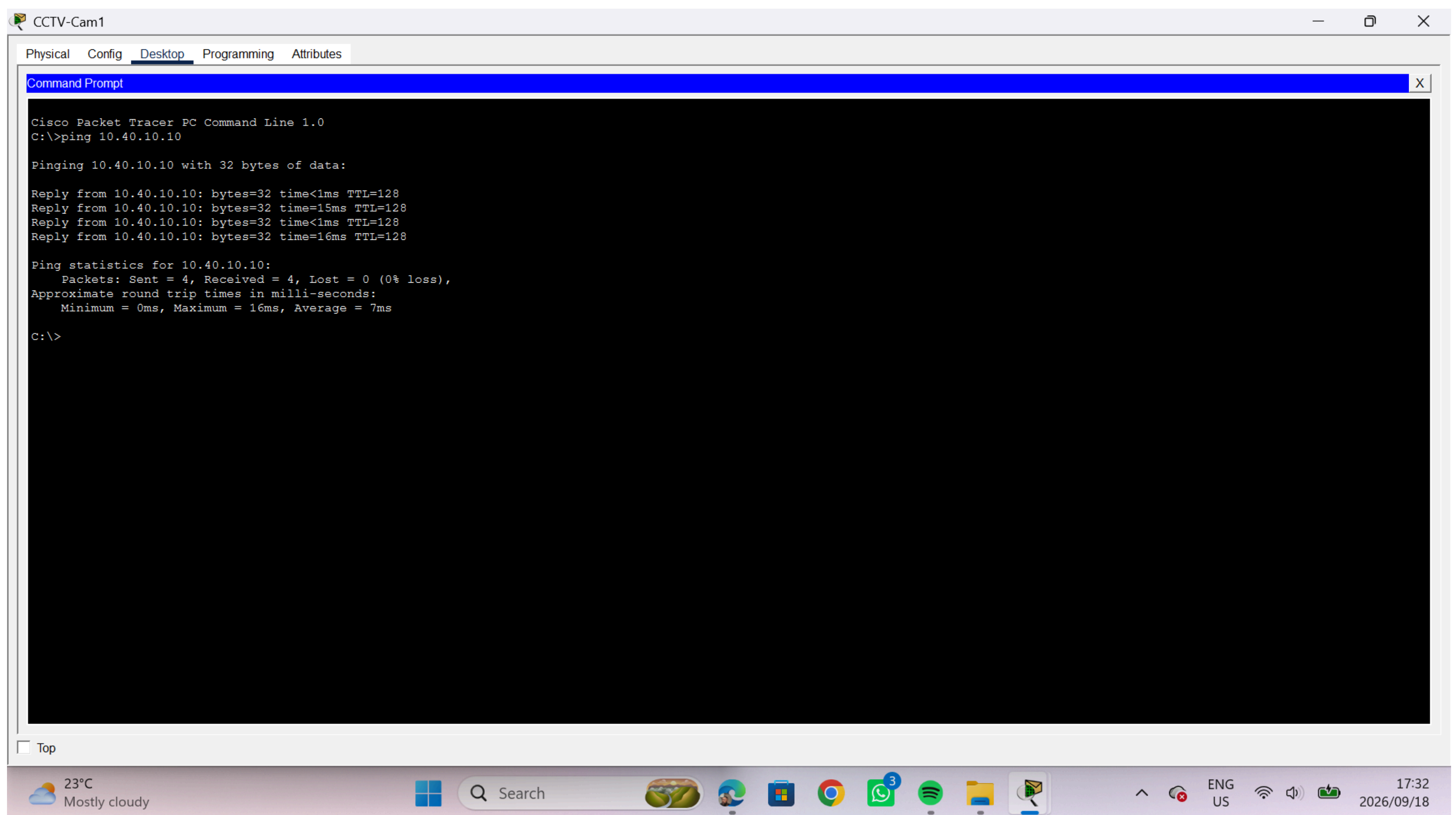
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128

Reply from 10.40.10.10: bytes=32 time<1ms TTL=128

Success rate is 100 percent (4/4)

Interpretation:  CCTV devices can communicate within VLAN 100 as expected. The ACL only blocks cross-VLAN traffic.

Evidence:



7.9 Test 7 — CCTV → Internet (No NAT for CCTV)

Source: CCTV-Cam1 (10.40.10.10)

Destination: 8.8.8.1

Command: ping 8.8.8.1

Expected: Blocked (CCTV excluded from NAT ACL)

Result:

Request timed out.

Request timed out.

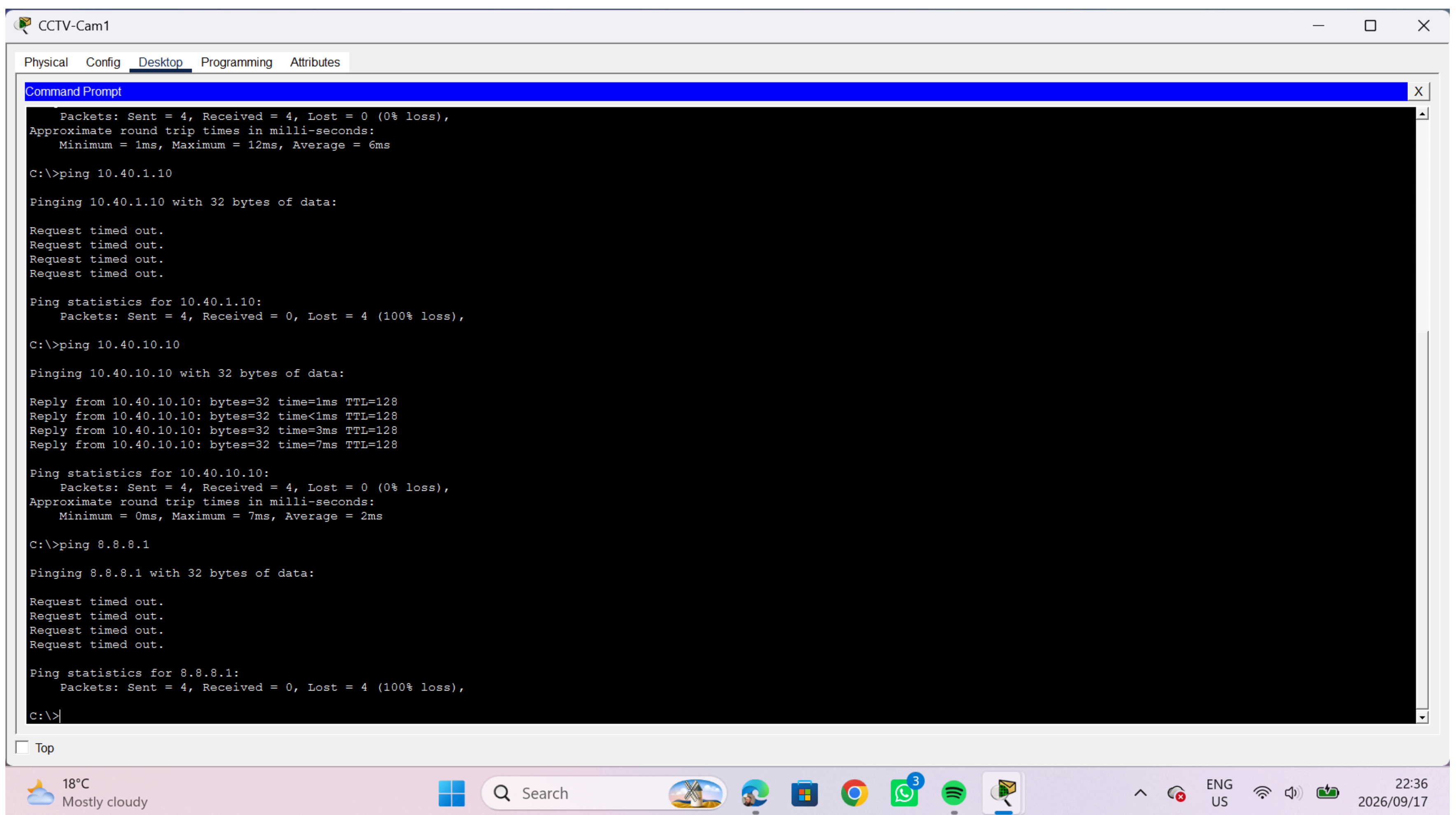
Request timed out.

Request timed out.

Success rate is 0 percent (0/4)

Interpretation: ☒ CCTV traffic cannot be translated to a public IP, so CCTV has no internet access. Enforces the segmentation constraint at the NAT layer.

Evidence:



7.10 Test 8 — NAT Translations Verification

Command: show ip nat translations (on Edge-Router)

Output:

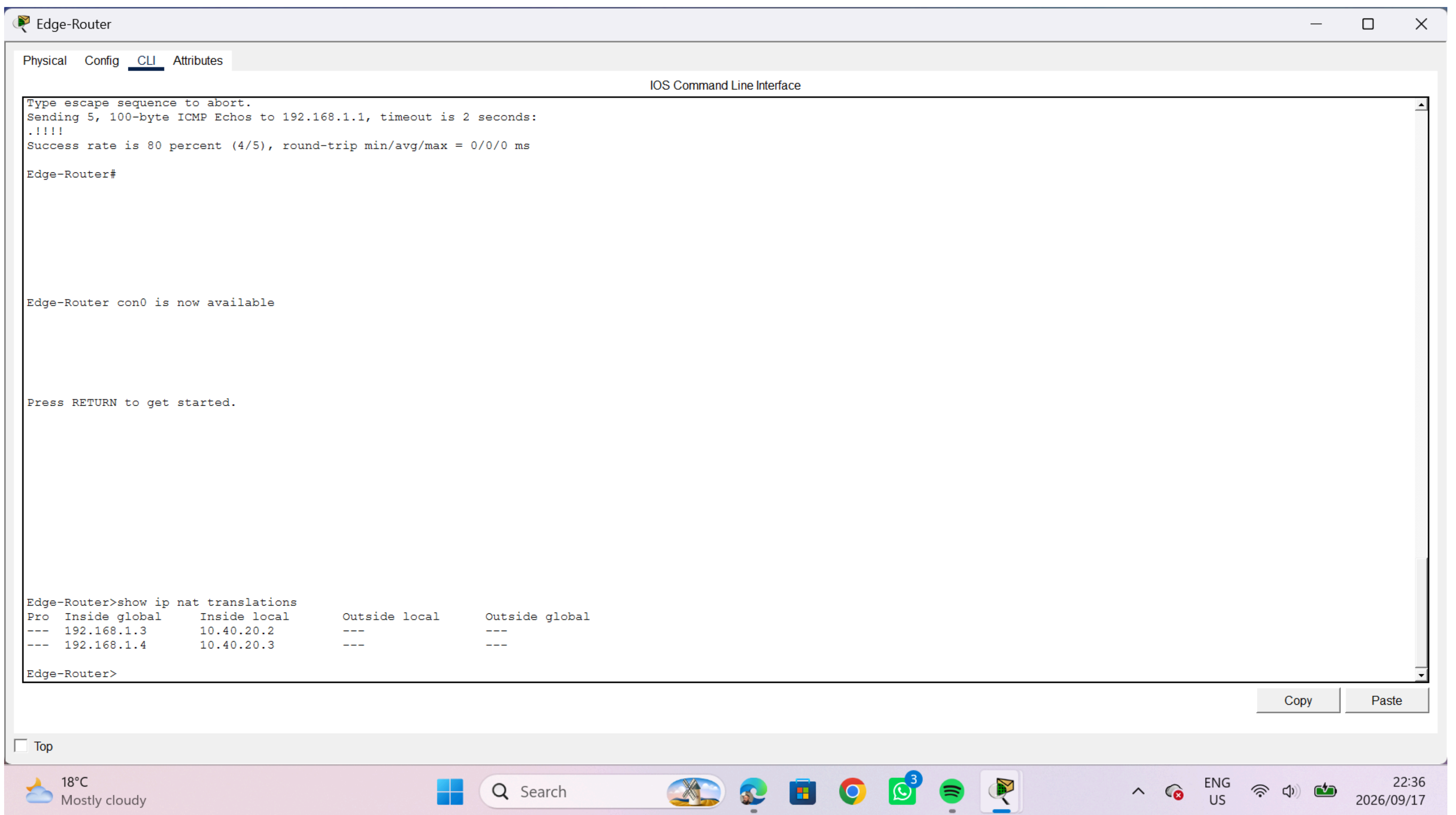
Pro Inside global Inside local Outside local Outside global

--- 192.168.1.3 10.40.20.2 --- ---

--- 192.168.1.4 10.40.20.3 --- ---

Interpretation:  Static NAT entries for App and File servers confirmed.

Evidence:



7.11 Test 9 — NAT Statistics

Command: show ip nat statistics (on Edge-Router)

Output:

Total translations: 2 (2 static, 0 dynamic, 0 extended)

Outside Interfaces: GigabitEthernet0/0/1

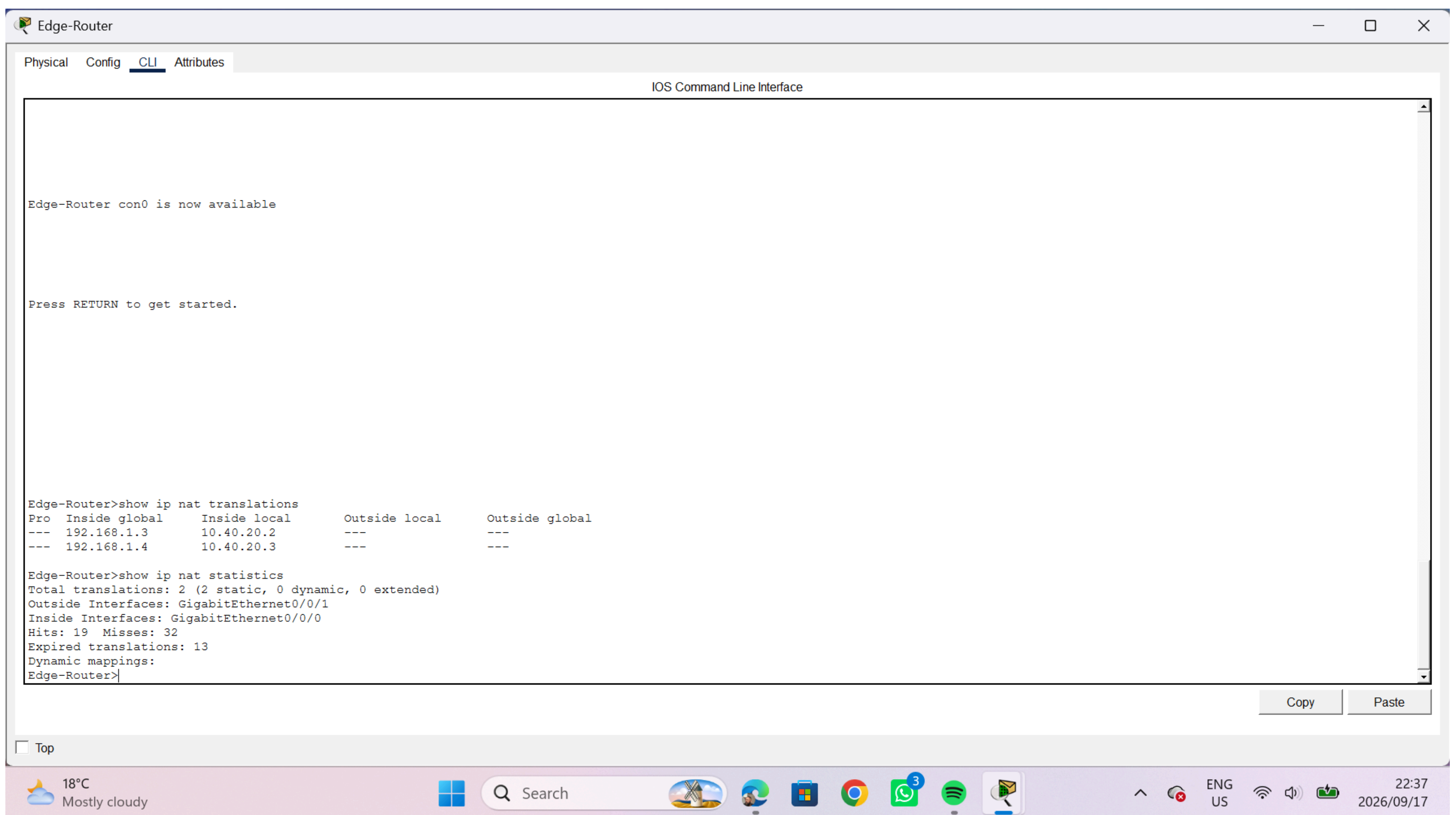
Inside Interfaces: GigabitEthernet0/0/0

Hits: 19 Misses: 32

Expired translations: 13

Interpretation: ☒ Inside and Outside interfaces correctly assigned. Hits confirm active use of NAT.

Evidence:



7.12 Test 10 — ACL Verification

Command: show ip access-lists (on Core-SW)

Output:

Extended IP access list 101

10 deny ip 10.40.10.0 0.0.0.255 10.40.0.0 0.0.255.255

20 permit ip any any

Extended IP access list 102

10 permit ip 10.40.1.0 0.0.0.255 10.40.20.0 0.0.0.15

20 permit ip 10.40.2.0 0.0.0.255 10.40.20.0 0.0.0.15

30 permit ip 10.40.5.0 0.0.0.255 10.40.20.0 0.0.0.15

40 permit ip 10.40.20.0 0.0.0.15 10.40.1.0 0.0.0.255 (4 match(es))

50 permit ip 10.40.20.0 0.0.0.15 10.40.2.0 0.0.0.255

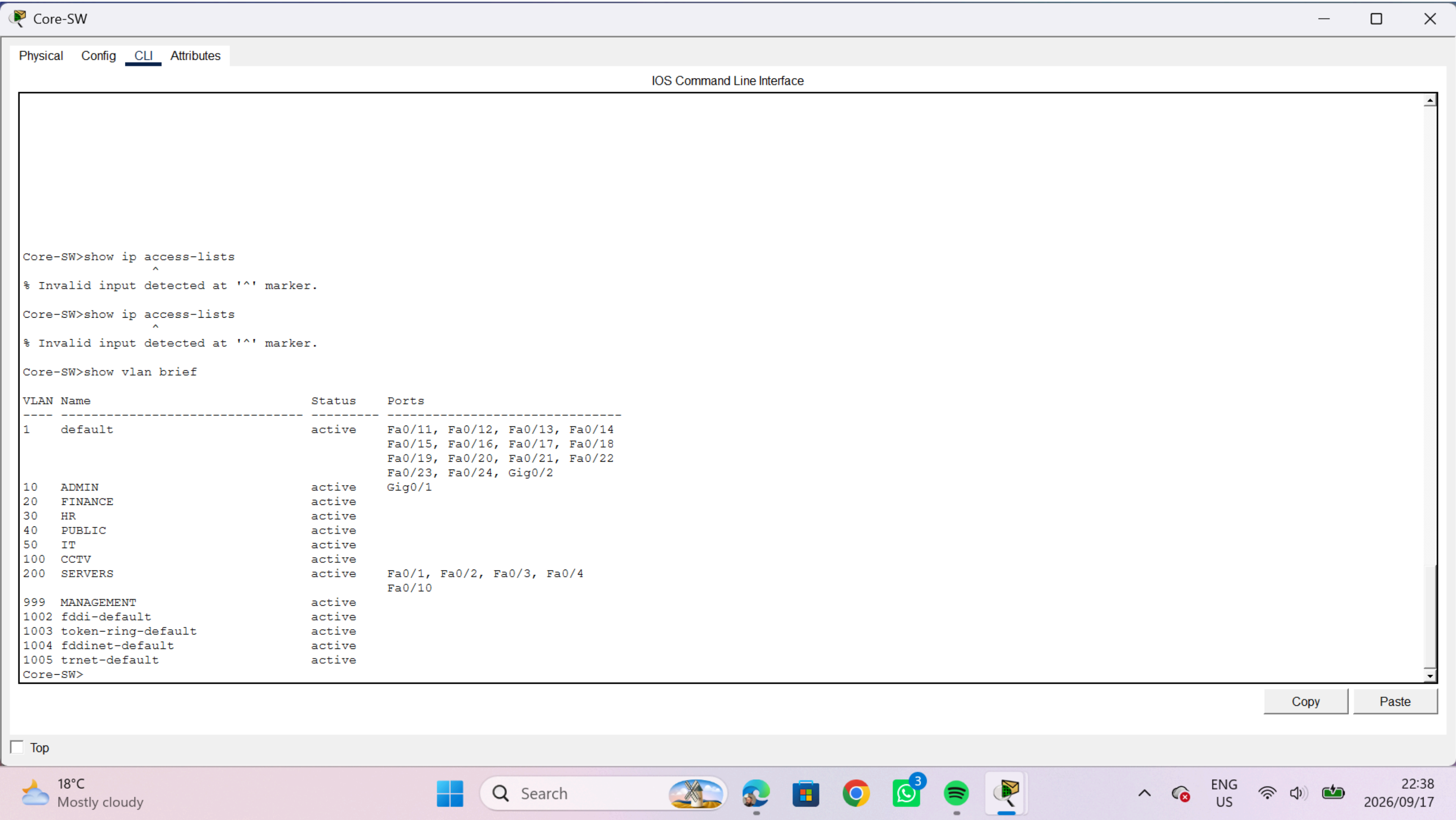
60 permit ip 10.40.20.0 0.0.0.15 10.40.5.0 0.0.0.255

70 permit ip 10.40.20.0 0.0.0.15 10.40.20.0 0.0.0.15

80 deny ip any any

Interpretation:  Both ACLs are correctly configured. Rule 40 shows 4 matches, confirming bidirectional traffic is being permitted between Admin and the servers.

Evidence:

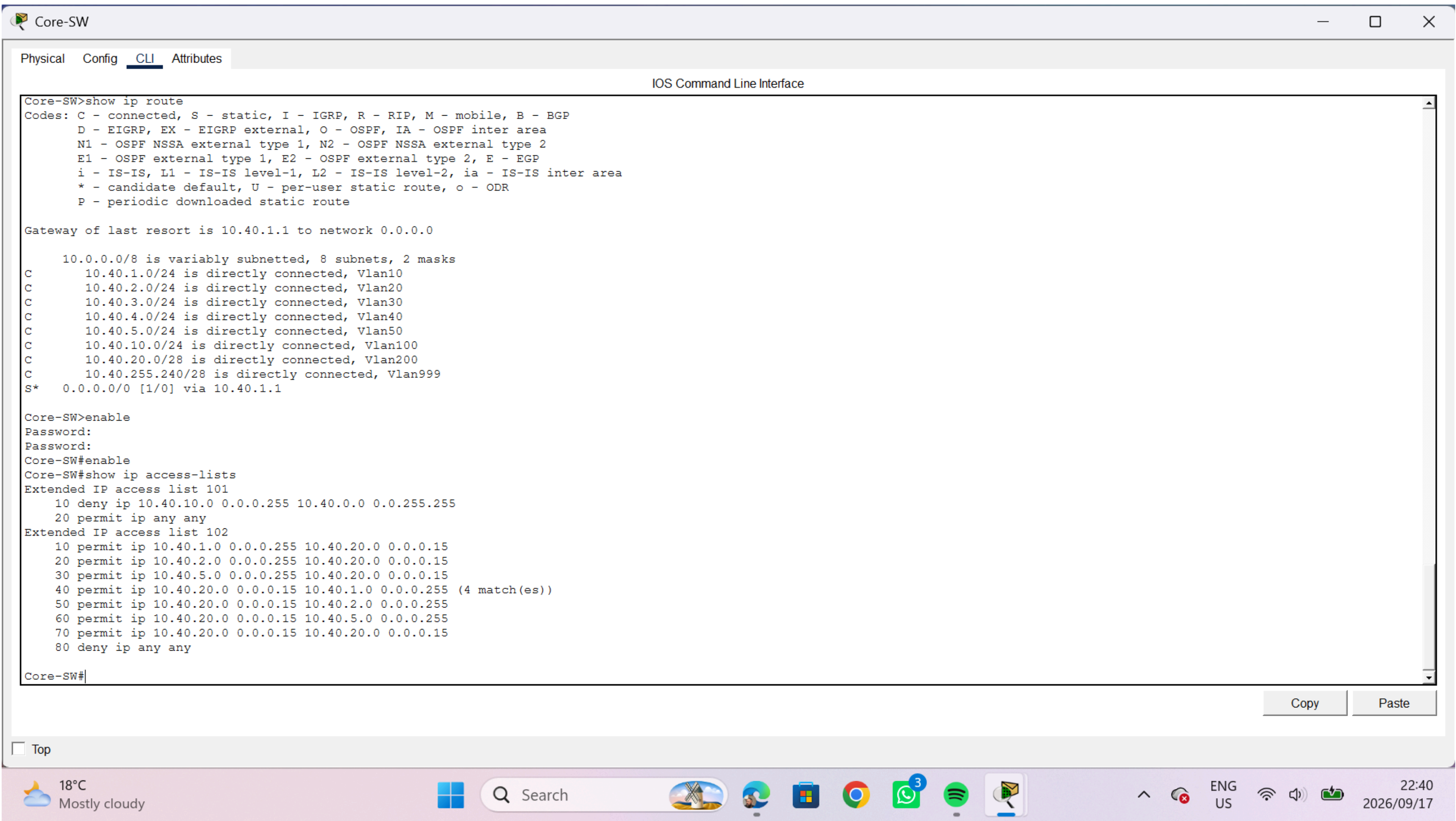


7.13 Test 11 — VLAN Configuration

Command: show vlan brief (on Core-SW)

Interpretation:  All 8 VLANs active with correct port assignments.

Evidence:



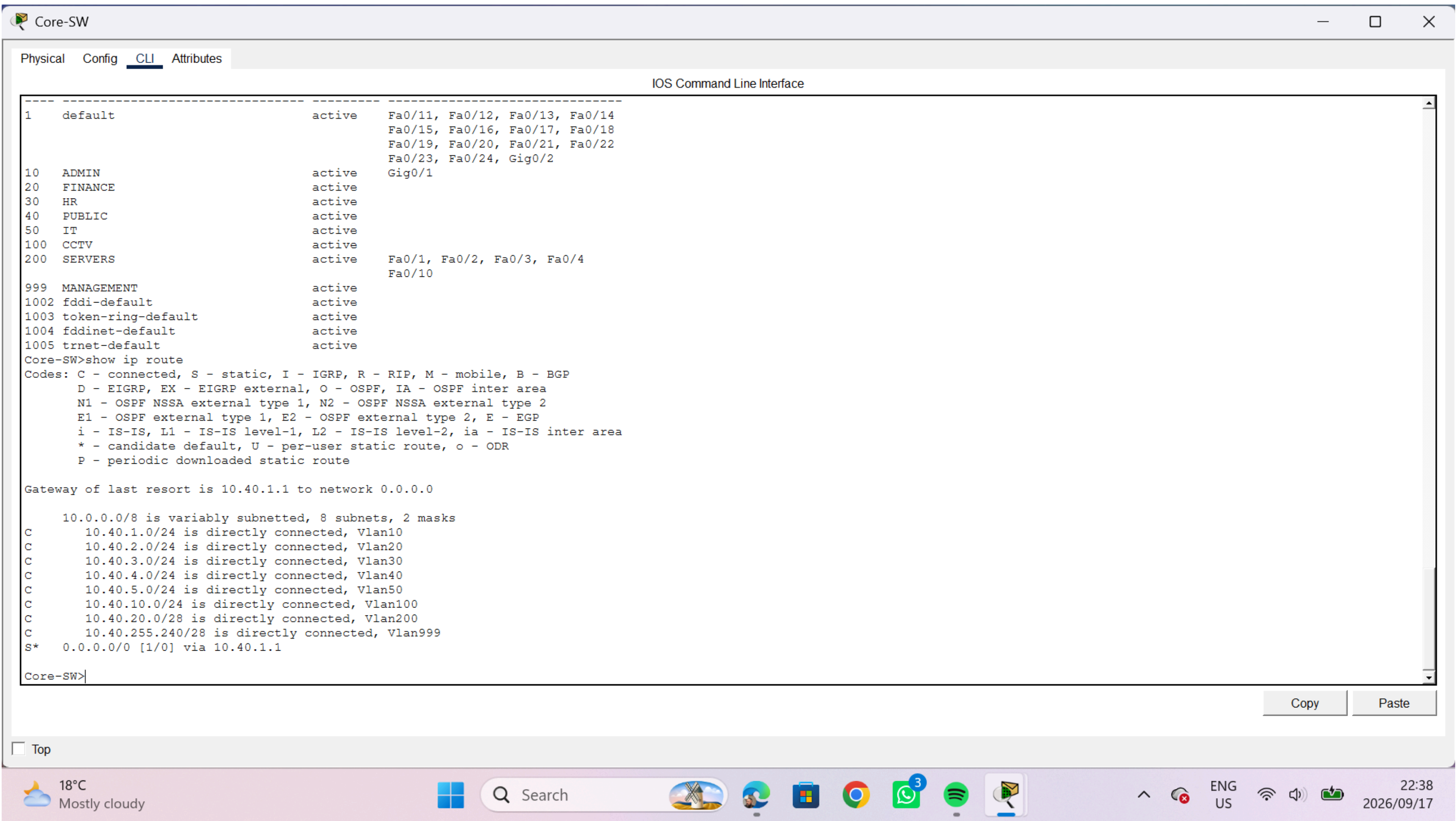
7.14 Test 12 — Routing Table

Command: show ip route (on Core-SW)

Output: 8 connected routes (SVIs) + 1 default route via 10.40.1.1

Interpretation: ☒ All VLANs are routed; default route points to Edge-Router.

Evidence:



7.15 Summary of Results

#	Test	Expected	Result
1	Admin → Finance	Success	✓ Pass
2	Admin → App-Server	Success	✓ Pass
3	HR → App-Server	Blocked	✓ Pass
4	HR → Internet	Success	✓ Pass
5	CCTV → Admin	Blocked	✓ Pass
6	CCTV → CCTV	Success	✓ Pass
7	CCTV → Internet	Blocked	✓ Pass
8	NAT Translations	2 static entries	✓ Pass
9	NAT Statistics	Interfaces assigned	✓ Pass
10	ACL Verification	Both ACLs present	✓ Pass
11	VLAN Configuration	8 VLANs active	✓ Pass

#	Test	Expected	Result
12	Routing Table	All SVIs + default	 Pass

All 12 tests passed. The network meets all specified requirements.

End of Section 7

8. TROUBLESHOOTING

8.1 Overview

During the design and implementation of the network, several technical challenges were encountered. This section documents the most significant issues and the troubleshooting steps taken to resolve them.

8.2 Issue 1 — DHCP Clients Received APIPA Addresses (169.254.x.x)

Problem

PCs in the Admin, Finance, HR, and Public Services VLANs could not obtain IP addresses via DHCP. They automatically assigned themselves APIPA addresses (169.254.x.x), indicating a failure to reach the DHCP server.

Root Cause

Access switch ports were not assigned to their departmental VLANs. Without VLAN tagging, DHCP requests stayed in VLAN 1 and never reached the DHCP service on the Core-SW.

Resolution

Assigned the correct VLAN to each access port:
interface range FastEthernet0/1-5
switchport mode access

```
switchport access vlan 10
spanning-tree portfast
```

Applied the same pattern on SW-Fin (VLAN 20), SW-HR (VLAN 30), SW-Public (VLAN 40), and SW-CCTV (VLAN 100).

Verification

After the fix, all PCs successfully obtained DHCP addresses on subsequent requests.

8.3 Issue 2 — Native VLAN Mismatch on Trunk Links

Problem

The CDP log on Core-SW showed repeated warnings:

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/9 (1),
with SW-CCTV FastEthernet0/1 (100).
```

Root Cause

The native VLAN on the trunk port did not match between Core-SW and SW-CCTV. This was a side effect of the port-configuration commands used earlier.

Resolution

Explicitly set the native VLAN to 1 on both ends:

On Core-SW:

```
interface FastEthernet0/9
switchport trunk native vlan 1
```

On SW-CCTV:

```
interface FastEthernet0/1
switchport trunk native vlan 1
```

Verification

The CDP warning disappeared after the configuration was saved on both ends.

8.4 Issue 3 — VLAN 200 SVI Used the Wrong /28 Subnet

Problem

Servers in VLAN 200 (10.40.20.0/28) could not communicate with the Core-SW, even though the physical links were up.

Root Cause

The SVI for VLAN 200 was initially configured as:

```
interface Vlan200
ip address 10.40.20.254 255.255.255.240
```

A /28 subnet spans 10.40.20.0 to 10.40.20.15 . The address 10.40.20.254 falls **outside this /28 range** (it would belong to 10.40.20.240/28). Servers with the mask /28 did not consider .254 to be on their subnet, so ARP requests never reached it.

Resolution

The SVI was reconfigured to the last usable IP in the correct /28 subnet:

```
interface Vlan200
ip address 10.40.20.14 255.255.255.240
```

Servers were also updated:

Server	IP	Gateway
App-Server	10.40.20.2	10.40.20.14
File-Server	10.40.20.3	10.40.20.14
DHCP-Server	10.40.20.4	10.40.20.14

Verification

Ping tests from Core-SW to each server succeeded after the correction. Server-to-SVI ping also succeeded.

8.5 Issue 4 — ACL 102 Blocked Return Traffic

Problem

Even after ACL 102 was applied, servers were unreachable. Admin-PC1 could not ping the App-Server.

Root Cause

Extended ACLs are **stateless** — they do not track connections. The initial ACL only permitted traffic **to** the server, not the reply **from** the server. Return packets were denied by the implicit `deny any any` at the end of the ACL.

Resolution

Added explicit bidirectional rules to ACL 102:

```
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.1.0 0.0.0.255
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.2.0 0.0.0.255
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.5.0 0.0.0.255
access-list 102 permit ip 10.40.20.0 0.0.0.15 10.40.20.0 0.0.0.15
```

Verification

Admin-PC1 could reach the App-Server with 4/4 packets after this change.

8.6 Issue 5 — ACLs Lost After Reload

Problem

During Milestone 3 preparation, verification tests showed that HR-PC1 could reach the App-Server (which should be blocked). The ACLs were no longer applied to the VLAN interfaces.

Root Cause

When the .pkt file was saved and reloaded between milestones, the ACL **interface assignments** were not preserved (the ACL rules themselves remained, but the `ip access-group` commands had to be re-applied).

Resolution

Re-applied the ACLs to the VLAN interfaces:

```
interface vlan 100
```

```
ip access-group 101 in
```

```
exit
```

```
interface vlan 200
```

```
ip access-group 102 in
```

```
exit
```

Verification

Subsequent tests confirmed HR was blocked from the server, and CCTV was blocked from office subnets.

Lesson Learned

Always verify `show running-config interface vlan X` after each milestone — to ensure ACLs are still attached to the correct interfaces.

8.7 Issue 6 — CCTV Devices Had No Internet Access (Desired)

Observation

CCTV pings to 8.8.8.1 returned 100% packet loss.

Root Cause

This is by design. The NAT ACL (access-list 1) does not permit `10.40.10.0/24`. CCTV traffic cannot be translated to a public IP.

Resolution

No action taken — this is the intended behaviour.

Interpretation

This behaviour explicitly enforces the CCTV segmentation constraint at the NAT layer. It is documented as a security feature rather than a bug.

8.8 Summary of Troubleshooting

#	Issue	Resolution	Status
1	DHCP failures	Assigned VLAN to access ports	✓
2	Native VLAN mismatch	Set native vlan 1 on both trunk ends	✓
3	VLAN 200 SVI subnet mismatch	Changed SVI IP to 10.40.20.14	✓
4	ACL 102 blocked replies	Added bidirectional permit rules	✓
5	ACLs lost after reload	Re-applied to VLAN interfaces	✓
6	CCTV no internet	By design — enforces constraint	✓

8.9 Key Lessons Learned

1. **Subnet masks matter** — a single misconfigured mask can break connectivity across an entire VLAN.
2. **Extended ACLs are stateless** — bidirectional rules are required when both request and reply must flow.
3. **Trunk consistency is critical** — native VLAN mismatches silently break inter-switch communication.
4. **Save-and-verify** — always verify critical config (ACLs, SVIs) after reloading a .pkt file.
5. **"Failures" aren't always failures** — some 100% packet loss results (CCTV → internet) prove that security policies are working.

End of Section 8

9. REFLECTION

9.1 Overall Reflection

Designing, implementing, and testing the Ramotshere Moiloa Local Municipality network has been a challenging but highly rewarding project. It combined theory (subnetting, VLANs, ACLs, NAT) with practical application in Cisco Packet Tracer and delivered a fully functional, service-rich enterprise network.

9.2 What I Learned

9.2.1 Technical Skills

- **VLSM and IP planning** — dividing a /16 block into functional subnets
- **Layer 2 segmentation** — VLAN design, trunk configuration, native VLAN consistency
- **Layer 3 routing** — SVIs, inter-VLAN routing, static routing, default routes
- **NAT** — understanding and configuring PAT (Overload) and Static NAT
- **ACL design** — extended ACLs, direction (in vs out), and stateless behaviour
- **DHCP** — configuring scope, gateway, DNS, and troubleshooting failures
- **Packet Tracer** — device selection, cabling, simulation, CLI configuration

9.2.2 Professional Skills

- **Documentation discipline** — using Markdown, screenshots, and a GitHub portfolio
- **Troubleshooting methodology** — systematic approach (isolate, verify, resolve)
- **Design justification** — explaining decisions in engineering terms
- **Change management** — handling CR4 within the existing design

9.2.3 Conceptual Understanding

- **Layered troubleshooting** — checking L1 (cables), L2 (VLANs), L3 (routing), L4+ (ACLs) in order
- **The value of "expected failures"** — packets dropped by ACLs are proof of working security policy
- **Trade-offs** — centralised DHCP vs. external, L3 core vs. router-on-a-stick, PAT vs. Static NAT
- **Defensive design** — planning for failures, verifying after each change

9.3 Challenges Faced

9.3.1 Subnet Mask Misconfiguration

The VLAN 200 SVI was initially configured with an IP outside its own /28 subnet. This caused hours of debugging before it was identified. Lesson: always re-check that a gateway IP falls within its subnet range.

9.3.2 Stateless ACLs

The initial ACL 102 only permitted traffic in one direction. This is a common conceptual trap. The fix — adding bidirectional permit rules — taught me how stateless ACLs actually operate in Cisco IOS.

9.3.3 Native VLAN Mismatch

The trunk between Core-SW and SW-CCTV silently dropped VLAN 100 traffic due to a native VLAN mismatch. This wasn't visible until the CDP log was checked. Lesson: always verify trunk configuration on both ends.

9.3.4 ACL Persistence

ACLs were found missing from interfaces after the .pkt was reloaded between milestones. This reinforced the importance of re-verifying configuration after each environment change.

9.4 What I Would Do Differently

Area	Improvement
Testing Order	Test each VLAN and ACL right after configuration, not at the end
Documentation	Document as I go, rather than reconstructing later
Backup Configs	Save configs to text files after each session
Version Control	Use Git commits more frequently for the .pkt file
Security	Add port security and DHCP snooping for extra hardening

9.5 Future Improvements

If the network were to be extended or productionised, the following improvements would be considered:

1. **Separate VLAN 50 for IT** — In this implementation, IT PCs share VLAN 10 with Admin. A future iteration would move IT to its own VLAN 50 subnet.
2. **Redundant links** — Implement a redundant uplink from Core-SW to Edge-Router, with HSRP or VRRP.
3. **DHCP Snooping** — Prevent rogue DHCP servers on the access layer.
4. **Port Security** — Limit MAC addresses per port to prevent unauthorised devices.
5. **Centralised Syslog Server** — For auditing and monitoring.
6. **NTP** — Sync timestamps across devices for accurate log correlation.
7. **AAA** — Implement TACACS+ or RADIUS for centralised authentication.
8. **Real Firewall** — Replace the Edge-Router's ACL with a dedicated firewall appliance for stateful inspection.
9. **VPN for Remote Access** — Allow secure remote connectivity for staff.
10. **Wi-Fi Integration** — Add wireless access points with WPA3, integrated into the same VLAN structure.

9.6 Personal Reflection

This project has given me a deep appreciation for how much invisible engineering goes into a functional network. What looks like "just an internet connection" is actually the result of careful VLAN planning, routing decisions, ACL rules, and NAT translations.

The most valuable lesson was learning to troubleshoot methodically. When HR-PC1 couldn't reach the App-Server despite "everything looking correct", we traced through L1 → L2 → L3 → L4+ in order, and found the problem was a stateless ACL blocking return traffic. That's the kind of insight only gained through hands-on experience.

I also appreciated the discipline required to document everything — not just the working parts, but the journey of fixing things. That's what separates a working project from a professional one.

9.7 Conclusion

The network successfully meets all project requirements:

-  Connectivity across all departments
-  Inter-VLAN routing
-  CCTV segmentation (constraint)
-  NAT (assigned challenge) — configured, verified, and demonstrated
-  CR4 server access control (change request)
-  DHCP services for office users
-  Fully documented, tested, and reproducible

I am confident this design would serve the Ramotshere Moiloa Local Municipality well — providing secure, reliable, and scalable connectivity that could be extended as the municipality grows.

End of Section 9

10. REFERENCES

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10.5 RFCs and Standards

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10.6 Referencing Style

All references follow the **APA 7th Edition** referencing style as specified by North-West University guidelines.

End of Section 10